

Homework 5

Definitions

Define the following terms/concepts

- a.) **Non-parametric statistics** - A branch of statistics which does not assume that the sample data comes from a specific probability distribution.
- b.) **Parametric statistics** - A branch of statistics which assumes the sample data comes from a specific probability distribution that depends on a set of fixed parameters.
- c.) Describe the advantages and disadvantages of **Non-parametric** significance tests

Advantages

1. They may be the only alternative when sample sizes are very small (unless the population distribution is known exactly, but this is almost never the case)
2. They make few assumptions about the population distribution of the data
3. They are advantages when the data represent crude measurements such as subjective ratings/rankings (e.g, Likert responses)
4. They often have simpler computations and interpretations than parametric tests

Disadvantages

1. They are generally less powerful than their parametric analogues
- d.) **Explanatory Variable** -In bivariate statistical procedures, the explanatory variable is the independent variable that usually defines the groups that will be compared across the response variable
 - e.) **Response Variable** -The dependent variable that measures the outcome of interest from each group

Confidence intervals and two-tailed tests:

(1.) For parts (a) - (d) you are given a confidence interval for the parameter along with the point estimate it was built from. Determine whether the null hypothesis H_0 should be **Rejected** or **Not Rejected** by checking whether the value claimed under H_0 falls inside the interval, and give the significance level α

a.) $H_0 : \mu = -2$ $H_A : \mu \neq -2$ 95% CI for $\mu \in [-3.0, -2.4]$ $\bar{x} = -2.7$

Decision = reject $H_0 \alpha = 0.05$

The null value -2 lies outside $[-3.0, -2.4]$, so H_0 is rejected at the 5% level.

b.) $H_0 : p = 0.9$ $H_A : p \neq 0.9$ 99% CI for $p \in [0.968, 0.972]$ $\hat{p} = 0.97$

Decision = reject $H_0 \alpha = 0.01$

The null value 0.9 lies outside $[0.968, 0.972]$, so H_0 is rejected at the 1% level.

c.) $H_0 : \mu = 0$ $H_A : \mu \neq 0$ 90% CI for $\mu \in [-3.63, 7.49]$ $\bar{x} = 1.93$

Decision = fail to reject H_0 $\alpha = 0.1$

The null value 0 lies inside $[-3.63, 7.49]$, so there is not enough evidence to reject H_0 at the 10% level.

d.) $H_0 : \mu = 50$ $H_A : \mu \neq 50$ 85% CI for $\mu \in [93.6, 139]$ $\bar{x} = 116.3$

Decision = reject H_0 $\alpha = 0.15$

The null value 50 lies outside $[93.6, 139]$, so H_0 is rejected at the 15% level.

The sign test

(2.) A dermatologist is conducting an experiment to test whether a new topical skin care cream reduces the appearance of facial acne. From a sample of 20 patients with acne the researcher gives each patient a score from 1 – 5, where (1) represents the mildest acne and (5) represents the most severe acne, before the topical cream is applied. Patients are then asked to use the topical cream for 10 days. Following the 10 day treatment, patients are given a new acne score between 1 – 5. The data for this experiment is given in the table below:

Patient Number	Score Before Treatment	Score After Treatment	Difference	Sign
1	3	2	1	+
2	2	1	1	+
3	4	1	3	+
4	4	1	3	+
5	4	1	3	+
6	5	2	3	+
7	1	1	0	
8	5	4	1	+
9	5	1	4	+
10	2	1	1	+
11	2	2	0	
12	5	1	4	+
13	3	3	0	
14	3	1	2	+
15	1	1	0	
16	2	1	1	+
17	4	1	3	+
18	5	1	4	+
19	2	3	-1	-
20	1	1	0	

The dermatologist is particularly interested in whether the new product improves the appearance of the patients' acne. They plan to test the null hypothesis $H_0 : p = 0.5$ using a right-tailed test with alternative hypothesis $H_A : p > 0.5$ using a sign test to determine if there is enough evidence to conclude that the topical cream has a positive effect on reducing facial acne at the $\alpha = 0.05$ level.

a.) Fill in the table above by computing the difference in acne scores before and after applying the skin care cream. Record the sign of each difference and report the total number of positive signs s . Note that

any patient whose score did not change (a difference of zero) is dropped from the test, so the number of observations used in the sign test is the number of non-zero differences.

Of the 20 patients, 14 had a positive difference, 1 had a negative difference, and 5 had a difference of zero. The five tied patients are discarded, leaving $n = 15$ observations with $s = 14$ positive signs.

b.) What does a “positive” sign indicate in this experiment? A positive sign indicates that the patient’s acne score went down after treatment - that is, the skin cream reduced the appearance of their acne

c.) Compute the p -value for the sign test The number of positive signs is $s = 14$ out of $n = 15$ non-zero differences. The p -value is the probability of observing 14 or more positive signs in 15 observations, given by

$$p\text{-value} = P(S \geq s | H_0) = \sum_{k=14}^{15} \frac{15!}{k!(15-k)!} \cdot 0.5^k \cdot (1-0.5)^{15-k}$$

$$\approx 0.0005$$

d.) Use the p -value from part (c) to determine if there is enough evidence to conclude that the topical cream product has a positive effect on reducing the appearance of acne. Interpret your decision in context.

Since the p -value 0.0005 is less than $\alpha = 0.05$, we reject the null hypothesis at the 5% significance level. There is sufficient evidence to conclude that the topical cream reduces the appearance of facial acne

(3.) In a boreal forest, researchers conducted an experiment to study how herbivores respond to variations in food plant quality and quantity at the stand level. They fertilized young forest stands and observed herbivore use over the subsequent year. The data collected included the number of animal tracks in the fertilized and control plots (Ball, Danell, and Sunesson 2000)

Observation	Number of Tracks In Fertilized plots	Number of Tracks In Control Plots	Difference	Sign
1	15	10	5	+
2	12	9	3	+
3	18	11	7	+
4	14	8	6	+
5	16	12	4	+
6	13	10	3	+
7	17	11	6	+
8	14	9	5	+
9	19	13	6	+
10	15	10	5	+
11	11	8	3	+
12	16	12	4	+

The researchers are interested in whether or not fertilizing stands increased herbivore activity. Conduct a sign test at the $\alpha = 0.05$ significance level to determine if fertilized plots have significantly more herbivore tracks than control plots

a.) Fill in the table above by computing the difference in animal tracks between the fertilized and control plots. Record the sign of each difference and report the total number of positive signs s

b.) What does a “positive” sign indicate in this experiment? A positive sign indicates that the fertilized stands have more animal tracks

c.) Compute the p -value for the sign test Every one of the 12 differences is positive, so $s = 12$ out of $n = 12$ (there are no ties to discard). The p -value is the probability of observing 12 or more positive signs in 12

observations, given by

$$\begin{aligned} p\text{-value} &= P(S \geq s | H_0) = \frac{12!}{12!(12-12)!} \cdot 0.5^{12} \cdot (1-0.5)^{12-12} \\ &= 0.5^{12} \\ &\approx 0.00024 \end{aligned}$$

d.) Use the p -value from part (c) to determine if there is enough evidence to conclude that fertilized stands have more herbivore activity. At the $\alpha = 0.05$ significance level we reject the null hypothesis and conclude that fertilized stands have significantly more animal activity than non-fertilized stands

Comparing two population proportions p_1 and p_2

(4.) The United States Supreme Court serves as the judicial branch of the U.S. government, tasked with ensuring that laws conform to the U.S. Constitution. Although traditionally operating as a relatively discreet arbiter within the American government, recent years have seen an unusual spotlight on the Supreme Court in partisan politics. This heightened attention is largely attributed by political analysts to several landmark Supreme Court decisions and ethical controversies that have surfaced. Consequently, it is believed that this increased scrutiny has adversely affected public approval of the Court across the political spectrum. The table below provides a summary of current and historical polling results for assessing SCOTUS job approval. The current approval rating is summarized from 192 political polls conducted among the public since late 2022 (Ryan and Bycoffe 2024). The historic approval sentiment is characterized from 39 Gallup polls conducted since early 2001.

Period	Parameter	Sample Size	Mean SCOTUS Approval Rating
Current Approval (2022 - 2024)	p_1	192	41.16
Past Approval (Gallup: 2001 - 2022)	p_2	39	50.87

Conduct a two-sample test at the $\alpha = 0.05$ significance level to determine whether the current approval rating is significantly lower than the historical average - be sure to address all five steps of the hypothesis test.

1. the pooled estimate of the common proportion is

$$\hat{p} = \frac{0.4116(192) + 0.5087(39)}{192 + 39} = 0.428$$

The success/failure condition must be checked in each group separately using the pooled $\hat{p}$: for group 1, $n_1\hat{p} = 192(0.428) = 82$ and $n_1(1 - \hat{p}) = 192(0.572) = 110$; for group 2, $n_2\hat{p} = 39(0.428) = 17$ and $n_2(1 - \hat{p}) = 39(0.572) = 22$. All four counts exceed the required 15, so the sample sizes are sufficient to meet the asymptotic requirement of the test

2. The null hypothesis is $H_0 : p_1 - p_2 = 0$ and the alternative hypothesis is $H_A : p_1 - p_2 < 0$
3. The pooled standard error is

$$SE_{D_{pooled}} = \sqrt{0.428(1 - 0.428) \left(\frac{1}{192} + \frac{1}{39} \right)} = 0.0869$$

the test statistic is

$$Z_{obs} = \frac{(0.4116 - 0.5087) - 0}{0.0869} = -1.11$$

4. The p -value is given by

$$p\text{-value} = P(Z \leq -1.11|H_0) = 0.132$$

Note that your p -value may be slightly different depending on how you rounded

5. At the $\alpha = 0.05$ significance level, we fail to reject the null hypothesis of no difference and conclude that the current SCOTUS approval rating is not significantly lower than the historical average of about 50.9% approving

(5.) A study published in the Journal of Paediatrics and Child Health was interested in the number of reports of child abuse in Sergipe, Brazil, before and after COVID-19 started. The researchers hypothesized that children in abusive homes might be in more danger because they weren't going to school and their families were under more stress (Martins-Filho et al. 2020). A summary of the study is reported below:

Period	Parameter	Total Registries	Registries for Children under 12 Years of Age	Relative Proportion
Before COVID-19 (2019)	p_1	70	18	
After COVID-19 (2020)	p_2	53	16	

Conduct a two-sample test at the $\alpha = 0.01$ significance level to determine whether the proportion of abuse registries involving children under the age of 12 was significantly different after the start of the COVID-19 pandemic - be sure to address all five steps of the hypothesis test. (Note that this proportion is the share of all registries that involve a child under 12, not the underlying rate of abuse in the population.)

1. the pooled estimate of the common proportion is

$$\hat{p} = \frac{18 + 16}{70 + 53} = 0.276$$

Checking the success/failure condition in each group using the pooled $\hat{p}$: for group 1, $n_1\hat{p} = 70(0.276) = 19.3$ and $n_1(1 - \hat{p}) = 70(0.724) = 50.7$; for group 2, $n_2\hat{p} = 53(0.276) = 14.6$ and $n_2(1 - \hat{p}) = 53(0.724) = 38.4$. Note that $n_2\hat{p} = 14.6$ falls just below the guideline of 15, so the normal approximation here is borderline and the results should be interpreted with some caution

2. The null hypothesis is $H_0 : p_1 - p_2 = 0$ and the alternative hypothesis is $H_A : p_1 - p_2 \neq 0$
 3. The pooled standard error is

$$SE_{D_{pooled}} = \sqrt{0.276(1 - 0.276) \left(\frac{1}{70} + \frac{1}{53} \right)} = 0.0819$$

the test statistic is

$$Z_{obs} = \frac{(0.26 - 0.30) - 0}{0.0819} = -0.488$$

4. The p -value is given by

$$\begin{aligned} p\text{-value} &= P(|Z| \geq | -0.488 | | H_0) \\ &= 2 \times [1 - P(Z < 0.488 | H_0)] = 0.625 \end{aligned}$$

Note that your p -value may be slightly different depending on how you rounded

5. At the $\alpha = 0.01$ significance level we fail to reject the null hypothesis and conclude that the proportion of abuse registries involving children under 12 years of age before and after the start of the COVID-19 pandemic are not significantly different

Comparing two population means μ_1 and μ_2

(6.) A study published in the journal Physical Culture and Sport. Studies and Research explored what motivates athletes to keep playing sports, comparing team sports like football with individual ones like taekwondo (Moradi, Bahrami, and Dana 2020). The study used stratified random sampling to survey 265 athletes from four team disciplines (football, volleyball, basketball, handball) and 2 individual disciplines (kung fu and taekwondo). The study evaluated motivational factors affecting sport participation based on eight fields: achievement/status, teamwork, fitness, energy release, situational factors, skill development, friendship, and fun. Each field was evaluated based on a survey in which every item was scored on a three-level likert scale (very important=3, somewhat important=2, and not important=1). A field contains several items, and an athlete's score for that field is the **sum** of their item responses - so field scores range well above 3. The results of the study are summarized in the table below:

Motivational Factor	Sport Type	Sample Mean	Sample Stdev.	Sample Size
Participation (total)	Team	75.67	9.38	203
	Individual	80.50	8.40	62
Achievement	Team	15.03	2.71	203
	Individual	15.72	3.03	62
Teamwork	Team	8.11	1.21	203
	Individual	7.68	1.23	62
Energy Release	Team	11.88	2.05	203
	Individual	12.85	2.07	62
Fitness	Team	7.79	1.27	203
	Individual	8.40	0.94	62
Situational Factors	Team	7.48	1.44	203
	Individual	8.33	0.98	62
Skill Development	Team	7.94	1.28	203
	Individual	8.42	1.07	62
Friendship	Team	9.57	1.78	203
	Individual	10.50	1.51	62
Fun	Team	7.74	1.20	203
	Individual	8.18	1.04	62

Conduct a two independent samples t -test at the $\alpha = 0.05$ significance level to determine whether fitness as a motivating factor is significantly different between team and individual sports. Assume the populations have the SAME variance.

1. The null hypothesis is $H_0 : \mu_1 - \mu_2 = 0$ where μ_1 is the mean score for fitness for team sports and μ_2 is the mean score for fitness for individual sports. The alternative hypothesis is $H_A : \mu_1 - \mu_2 \neq 0$
2. The pooled estimate of the sample variance is

$$s_{pooled} = \sqrt{\frac{(203 - 1)1.27^2 + (62 - 1)0.94^2}{203 + 62 - 2}} = 1.201$$

the estimated standard error assuming the samples are independent and have the same variance is given by

$$SE(\hat{\mu}_d) = 1.201 \cdot \sqrt{\frac{1}{203} + \frac{1}{62}} = 0.174$$

The test statistic is

$$t_{obs} = \frac{(7.79 - 8.40) - 0}{0.174} = -3.505$$

where t_{obs} is approximately t -distributed with $203 + 62 - 2 = 263$ degrees of freedom

3. The p -value is

$$\begin{aligned} p\text{-value} &= P(|t| \geq |t_{obs}| | H_0) \\ &= 2 [P(t \geq 3.505 | H_0)] = 0.00054 \end{aligned}$$

4. At the $\alpha = 0.05$ significance level, we reject the null hypothesis and conclude that fitness as motivator is significantly different for athletes in team sports vs individual sports.

(7.) A 1993 study published in the Canadian Journal of Zoology investigated the characteristics of maturation and growth in two lowland populations (Servotte and Thevenon) of European Common Frogs *Rana temporaria* (Augert and Joly 1993) in Southeast France from 1986 - 1989. A summary of the body lengths for male and female frogs from 1 - 4 years in age in both populations are summarized below.

Age (Years)	Population Location	Male Body Length (mm)	Stdev	Sample size	Female Body Length (mm)	Stdev
1	Servotte	32.5	5.3	45	33.4	5.3
2	Servotte	55.8	4.7	11	59.7	6.1
3	Servotte	61.2	5.3	35	65.6	5.3
4	Servotte	64.7	7.3	12	69.6	0.7
1	Thevenon	32.9	3.9	49	32.8	4.1
2	Thevenon	51.8	3.2	26	53.3	6.1
3	Thevenon	59.9	5.4	27	59.9	6.1
4	Thevenon	61.0	4.1	12	63.0	4.1

Conduct a two independent samples t -test at the $\alpha = 0.05$ significance level to evaluate whether there is a statistically significant difference in the mean body lengths of four-year-old male frogs between the Servotte and Thévenon populations. Assume the populations have DIFFERENT variances.

1. The null hypothesis is $H_0 : \mu_1 - \mu_2 = 0$ where μ_1 is the mean male body length for frogs in the Servotte population and μ_2 is the mean male body length for frogs in the Thévenon population. The alternative hypothesis is $H_A : \mu_1 - \mu_2 \neq 0$
2. The estimated standard error assuming the samples are independent and do not have the same variance is given by

$$SE(\hat{\mu}_d) = \sqrt{\frac{7.3^2}{12} + \frac{4.1^2}{12}} = 2.417$$

The test statistic is

$$t_{obs} = \frac{(64.7 - 61.0)}{2.417} = 1.531$$

where t_{obs} is approximately t -distributed with 11 degrees of freedom

3. The p -value is

$$\begin{aligned} p\text{-value} &= P(|t| \geq |t_{obs}| | H_0) \\ &= 2 [P(t \geq 1.531 | H_0)] = 2(0.077) = 0.154 \end{aligned}$$

4. At the $\alpha = 0.05$ significance level, we fail to reject the null hypothesis and conclude that body lengths of four year old male frogs is not significantly different between the two populations.

References

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